

# Tutorial Window Function con DuckDB (2ª parte)

👉 [Enlace 1ª parte](#)

```
pip install duckdb
```

```
import duckdb

%load_ext sql
%config SqlMagic.autopandas = True
%config SqlMagic.displaycon = False

conn = duckdb.connect('BBDD/WindowFunction.duckdb')
%sql conn --alias duckdb
```

```
# con.sql("show databases")
conn.sql("show tables")
```

|             |
|-------------|
| name        |
| varchar     |
| order_items |
| orders      |
| products    |

## GROUP BY VS PARTITION BY

```
%sql SELECT model_year, AVG(list_price) avg_price\
      FROM products\
      GROUP BY model_year
```

|   | model_year | avg_price   |
|---|------------|-------------|
| 0 | 2016       | 980.299231  |
| 1 | 2017       | 1279.931176 |
| 2 | 2018       | 1658.470441 |
| 3 | 2019       | 2583.323333 |

```
%sql SELECT model_year, product_name, list_price,\
      AVG(list_price) OVER (PARTITION BY model_year) avg_price\
```

```
FROM products\
ORDER BY product_name
```

|     | model_year | product_name                                     | list_price | avg_price   |
|-----|------------|--------------------------------------------------|------------|-------------|
| 0   | 2018       | Electra Amsterdam Fashion 3i Ladies' - 2017/2018 | 899.99     | 1658.470441 |
| 1   | 2017       | Electra Amsterdam Fashion 7i Ladies' - 2017      | 1099.99    | 1279.931176 |
| 2   | 2017       | Electra Amsterdam Original 3i - 2015/2017        | 659.99     | 1279.931176 |
| 3   | 2017       | Electra Amsterdam Original 3i Ladies' - 2017     | 659.99     | 1279.931176 |
| 4   | 2018       | Electra Amsterdam Royal 8i - 2017/2018           | 1259.90    | 1658.470441 |
| ... | ...        | ...                                              | ...        | ...         |
| 316 | 2017       | Trek X-Caliber 8 - 2017                          | 999.99     | 1279.931176 |
| 317 | 2018       | Trek X-Caliber 8 - 2018                          | 999.99     | 1658.470441 |
| 318 | 2018       | Trek X-Caliber Frameset - 2018                   | 1499.99    | 1658.470441 |
| 319 | 2018       | Trek XM700+ - 2018                               | 3499.99    | 1658.470441 |
| 320 | 2018       | Trek XM700+ Lowstep - 2018                       | 3499.99    | 1658.470441 |

## COUNT DISTINCT

```
%%sql
SELECT year(order_date) y, COUNT(DISTINCT order_id) num_orders
FROM orders
GROUP BY year(order_date)
```

|   | y    | num_orders |
|---|------|------------|
| 0 | 2017 | 688        |
| 1 | 2016 | 635        |
| 2 | 2018 | 292        |

## RANK()

```
%%sql
/* Rank price from LOW->HIGH */
SELECT product_name, list_price,
       RANK() OVER (ORDER BY list_price ASC) rank
FROM products
```

|   | product_name                           | list_price | rank |
|---|----------------------------------------|------------|------|
| 0 | Strider Classic 12 Balance Bike - 2018 | 89.99      | 1    |
| 1 | Sun Bicycles Lil Kitt'n - 2017         | 109.99     | 2    |

|     | product_name                           | list_price | rank |
|-----|----------------------------------------|------------|------|
| 2   | Trek Boy's Kickster - 2015/2017        | 149.99     | 3    |
| 3   | Trek Girl's Kickster - 2017            | 149.99     | 3    |
| 4   | Trek Kickster - 2018                   | 159.99     | 5    |
| ... | ...                                    | ...        | ...  |
| 316 | Trek Domane SL Frameset - 2018         | 6499.99    | 316  |
| 317 | Trek Domane SL Frameset Women's - 2018 | 6499.99    | 316  |
| 318 | Trek Emonda SLR 8 - 2018               | 6499.99    | 316  |
| 319 | Trek Domane SLR 8 Disc - 2018          | 7499.99    | 320  |
| 320 | Trek Domane SLR 9 Disc - 2018          | 11999.99   | 321  |

```

%%sql
/* Rank price from HIGH->LOW */
SELECT product_name, list_price,
       RANK() OVER (ORDER BY list_price DESC) rank
FROM products

```

|     | product_name                           | list_price | rank |
|-----|----------------------------------------|------------|------|
| 0   | Trek Domane SLR 9 Disc - 2018          | 11999.99   | 1    |
| 1   | Trek Domane SLR 8 Disc - 2018          | 7499.99    | 2    |
| 2   | Trek Silque SLR 8 Women's - 2017       | 6499.99    | 3    |
| 3   | Trek Domane SL Frameset - 2018         | 6499.99    | 3    |
| 4   | Trek Domane SL Frameset Women's - 2018 | 6499.99    | 3    |
| ... | ...                                    | ...        | ...  |
| 316 | Trek Kickster - 2018                   | 159.99     | 317  |
| 317 | Trek Boy's Kickster - 2015/2017        | 149.99     | 318  |
| 318 | Trek Girl's Kickster - 2017            | 149.99     | 318  |
| 319 | Sun Bicycles Lil Kitt'n - 2017         | 109.99     | 320  |
| 320 | Strider Classic 12 Balance Bike - 2018 | 89.99      | 321  |

## FIRST\_VALUE

```

%%sql
/* Find the difference of price from cheapest alternative */
SELECT
  product_name,
  list_price,
  FIRST_VALUE(list_price) OVER (ORDER BY list_price ROWS BETWEEN UNBOUNDED
PRECEDING AND UNBOUNDED FOLLOWING ) cheapest_price,
  list_price - FIRST_VALUE(list_price) OVER (ORDER BY list_price ROWS BETWEEN

```

```
UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING) diff
FROM products
```

|     | product_name                           | list_price | cheapest_price | diff    |
|-----|----------------------------------------|------------|----------------|---------|
| 0   | Strider Classic 12 Balance Bike - 2018 | 89.99      | 89.99          | 0.0     |
| 1   | Sun Bicycles Lil Kitt'n - 2017         | 109.99     | 89.99          | 20.0    |
| 2   | Trek Boy's Kickster - 2015/2017        | 149.99     | 89.99          | 60.0    |
| 3   | Trek Girl's Kickster - 2017            | 149.99     | 89.99          | 60.0    |
| 4   | Trek Kickster - 2018                   | 159.99     | 89.99          | 70.0    |
| ... | ...                                    | ...        | ...            | ...     |
| 316 | Trek Domane SL Frameset - 2018         | 6499.99    | 89.99          | 6410.0  |
| 317 | Trek Domane SL Frameset Women's - 2018 | 6499.99    | 89.99          | 6410.0  |
| 318 | Trek Emonda SLR 8 - 2018               | 6499.99    | 89.99          | 6410.0  |
| 319 | Trek Domane SLR 8 Disc - 2018          | 7499.99    | 89.99          | 7410.0  |
| 320 | Trek Domane SLR 9 Disc - 2018          | 11999.99   | 89.99          | 11910.0 |

## LAST\_VALUE

```
%%sql
/* Find the difference of price from the priciest alternative */
SELECT
    product_name,
    list_price,
    LAST_VALUE(list_price) OVER (ORDER BY list_price ROWS BETWEEN UNBOUNDED
PRECEDING AND UNBOUNDED FOLLOWING ) highest_price,
    LAST_VALUE(list_price) OVER (ORDER BY list_price ROWS BETWEEN UNBOUNDED
PRECEDING AND UNBOUNDED FOLLOWING ) - list_price diff
FROM products
```

|     | product_name                           | list_price | highest_price | diff    |
|-----|----------------------------------------|------------|---------------|---------|
| 0   | Strider Classic 12 Balance Bike - 2018 | 89.99      | 11999.99      | 11910.0 |
| 1   | Sun Bicycles Lil Kitt'n - 2017         | 109.99     | 11999.99      | 11890.0 |
| 2   | Trek Boy's Kickster - 2015/2017        | 149.99     | 11999.99      | 11850.0 |
| 3   | Trek Girl's Kickster - 2017            | 149.99     | 11999.99      | 11850.0 |
| 4   | Trek Kickster - 2018                   | 159.99     | 11999.99      | 11840.0 |
| ... | ...                                    | ...        | ...           | ...     |
| 316 | Trek Domane SL Frameset - 2018         | 6499.99    | 11999.99      | 5500.0  |
| 317 | Trek Domane SL Frameset Women's - 2018 | 6499.99    | 11999.99      | 5500.0  |
| 318 | Trek Emonda SLR 8 - 2018               | 6499.99    | 11999.99      | 5500.0  |
| 319 | Trek Domane SLR 8 Disc - 2018          | 7499.99    | 11999.99      | 4500.0  |

|     | product_name                  | list_price | highest_price | diff |
|-----|-------------------------------|------------|---------------|------|
| 320 | Trek Domane SLR 9 Disc - 2018 | 11999.99   | 11999.99      | 0.0  |

## LAG

```

%%sql
/* Find number of orders in a year */
WITH yearly_orders AS (
  SELECT
    year(order_date) y,
    COUNT(DISTINCT order_id) num_orders
  FROM orders
  GROUP BY year(order_date)
)

/* Compare this year's sales to last year's */
SELECT
  *,
  LAG(num_orders) OVER (ORDER BY y) last_year_order,
  LAG(num_orders) OVER (ORDER BY y) - num_orders diff_from_last_year
FROM yearly_orders

```

|   | y    | num_orders | last_year_order | diff_from_last_year |
|---|------|------------|-----------------|---------------------|
| 0 | 2016 | 635        | <NA>            | <NA>                |
| 1 | 2017 | 688        | 635             | -53                 |
| 2 | 2018 | 292        | 688             | 396                 |

## LEAD

```

%%sql
/* Find number of orders in a year */
WITH yearly_orders AS (
  SELECT
    year(order_date) y,
    COUNT(DISTINCT order_id) num_orders
  FROM orders
  GROUP BY year(order_date)
)

/* Compare number of years compared to next year */
SELECT *,
  LEAD(num_orders) OVER (ORDER BY y) next_year_order,
  LEAD(num_orders) OVER (ORDER BY y) - num_orders diff_from_next_year
FROM yearly_orders

```

|   | y    | num_orders | next_year_order | diff_from_next_year |
|---|------|------------|-----------------|---------------------|
| 0 | 2016 | 635        | 688             | 53                  |

|   | y    | num_orders | next_year_order | diff_from_next_year |
|---|------|------------|-----------------|---------------------|
| 1 | 2017 | 688        | 292             | -396                |
| 2 | 2018 | 292        | <NA>            | <NA>                |

## RANKING

DENSE\_RANK(), ROW\_NUMBER(), RANK(), PERCENT\_RANK(), NTILE(),  
CUME\_DIST()

```

%%sql
/* Rank all products by price */
SELECT
product_name,
list_price,
ROW_NUMBER() OVER (ORDER BY list_price) row_num,
DENSE_RANK() OVER (ORDER BY list_price) dense_rank,
RANK() OVER (ORDER BY list_price) rank,
PERCENT_RANK() OVER (ORDER BY list_price) pct_rank,
NTILE(75) OVER (ORDER BY list_price) ntile,
CUME_DIST() OVER (ORDER BY list_price) cume_dist
FROM products

```

|     | product_name                           | list_price | row_num | dense_rank | rank | pct_rank | ntile | cume_dist |
|-----|----------------------------------------|------------|---------|------------|------|----------|-------|-----------|
| 0   | Strider Classic 12 Balance Bike - 2018 | 89.99      | 1       | 1          | 1    | 0.000000 | 1     | 0.003115  |
| 1   | Sun Bicycles Lil Kitt'n - 2017         | 109.99     | 2       | 2          | 2    | 0.003125 | 1     | 0.006231  |
| 2   | Trek Boy's Kickster - 2015/2017        | 149.99     | 3       | 3          | 3    | 0.006250 | 1     | 0.012461  |
| 3   | Trek Girl's Kickster - 2017            | 149.99     | 4       | 3          | 3    | 0.006250 | 1     | 0.012461  |
| 4   | Trek Kickster - 2018                   | 159.99     | 5       | 4          | 5    | 0.012500 | 1     | 0.015576  |
| ... | ...                                    | ...        | ...     | ...        | ...  | ...      | ...   | ...       |
| 316 | Trek Domane SL Frameset - 2018         | 6499.99    | 317     | 104        | 316  | 0.984375 | 74    | 0.993769  |
| 317 | Trek Domane SL Frameset Women's - 2018 | 6499.99    | 318     | 104        | 316  | 0.984375 | 75    | 0.993769  |
| 318 | Trek Emonda SLR 8 - 2018               | 6499.99    | 319     | 104        | 316  | 0.984375 | 75    | 0.993769  |
| 319 | Trek Domane SLR 8 Disc - 2018          | 7499.99    | 320     | 105        | 320  | 0.996875 | 75    | 0.996885  |
| 320 | Trek Domane SLR 9 Disc - 2018          | 11999.99   | 321     | 106        | 321  | 1.000000 | 75    | 1.000000  |

## AVG, MIN, MAX

```
%%sql
SELECT
    order_id,
    product_id,
    discount,
    AVG(discount) OVER (PARTITION BY product_id) avg_discount,
    MIN(discount) OVER (PARTITION BY product_id) min_discount,
    MAX(discount) OVER (PARTITION BY product_id) max_discount
FROM order_items
WHERE order_id != 1
ORDER BY order_id
```

|      | order_id | product_id | discount | avg_discount | min_discount | max_discount |
|------|----------|------------|----------|--------------|--------------|--------------|
| 0    | 2        | 20         | 0.07     | 0.112530     | 0.05         | 0.20         |
| 1    | 2        | 16         | 0.05     | 0.113191     | 0.05         | 0.20         |
| 2    | 3        | 20         | 0.05     | 0.112530     | 0.05         | 0.20         |
| 3    | 3        | 3          | 0.05     | 0.105581     | 0.05         | 0.20         |
| 4    | 4        | 2          | 0.10     | 0.107792     | 0.05         | 0.20         |
| ...  | ...      | ...        | ...      | ...          | ...          | ...          |
| 4712 | 1614     | 124        | 0.07     | 0.125714     | 0.07         | 0.20         |
| 4713 | 1614     | 159        | 0.07     | 0.062000     | 0.05         | 0.07         |
| 4714 | 1615     | 214        | 0.07     | 0.085000     | 0.07         | 0.10         |
| 4715 | 1615     | 197        | 0.20     | 0.175000     | 0.10         | 0.20         |
| 4716 | 1615     | 182        | 0.20     | 0.130000     | 0.05         | 0.20         |

```
# Explicitly close the connection
conn.close()
```